Code No: 136ER **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, May - 2019
ARTIFICIAL NEURAL NETWORKS
(Common to EEE, CSE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART- A

(25 Marks)

1.a)	What is a Neural Network?	[2]
b)	Define Knowledge Representation technique.	[3]
c)	What is Linear Least Square Filters?	[2]
d)	What do you mean by Adaptive Filtering Problem?	[3]
e)	Define Back Propagation method.	[2]
f)	What is Supervised Learning?	[3]
g)	State SOM algorithm.	[2]
h)	What are properties of Feature Map?	[3]
i)	What is Computer Experiment?	[2]
j)	Define Dynamical Systems	[3]

PART-B

(50 Marks)

2. What are the various active building blocks of neural networks? Explain the current mirror and inverter based neuron in detail. [10]

OR

3.a) Explain about Memory Based Learning and Hebbian Learning.
b) Describe about Neural networks viewed as Directed Graphs. [5+5]

4.a) Write in detailed note on Unconstrained Organization Techniques and Learning curves.
b) Explain in detail Perceptron -Convergence theorem. [5+5]

OR

5.a) Discuss about solution for XOR problem using Back Propagation Algorithm.
b) Define Heuristics? Explain about Output Representation and decision Rule. [5+5]

6.a) Explain about Cross Validation and What is the purpose of Cross Validation.
b) Explain about Hessian Matrix in detail. [5+5]

OR

7.a) What are Network Pruning Techniques? Explain them in detail.
b) List and explain Virtues and Limitations of Back Propagation Learning. [5+5]

8.a) Explain Self Organization Map.
b) List and explain any two basic Feature Mapping Models. [5+5]

OR

9.a) Discuss Computer Simulations and Learning Vector Quantization.
b) Describe about Adaptive Patter Classification. [5+5]

10.a) Discuss about Stability of Equilibrium States and Attractors.
b) List and explain Neuro Dynamical models. [5+5]

OR

11.a) Write about Hopfield Models in detail.
b) Explain about Manipulation of Attractors as a Recurrent Network Paradigm. [5+5]

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